

Results

The glucose starch solution in the dialysis bag turned dark red and the glucose results for the solution decreased. The potassium iodide solution in the cup did not change color and maintained the same glucose results. The solution with 0.0M sucrose had the lowest change in mass and percent change in mass. Then 0.2M sucrose solution was the second lowest, then the 0.4M sucrose solution, then 0.6M sucrose solution, and then 0.8M sucrose solution. The 1.0M solution had the highest change in mass and percent change in mass. Table 1 displays the location, solution, initial and final solution colors, and initial and final glucose test results. Table 2 displays the sucrose solution concentrations, initial mass, final mass, change in mass, and percent change in mass.

Table 1: Solution Color and Glucose Test Results for Glucose Solution in Dialysis Bag and Potassium Iodide Solution in a Cup

Location	Solution	Initial Solution Color	Final Solution Color	Initial Glucose Test	Final Glucose Test
Dialysis Bag	15 % Glucose / 1 % Starch	Transparent (no color)	Dark Red	Brown 2000 mmol/L	Light Brown 1000 mmol/L
Cup	Potassium Iodide	Red/Orange	Red/Orange	Light Green 100 mmol/L	Light Green 100 mmol/L

Table 2: Initial Mass, Final Mass, Change in Mass, and Percent Change in Mass for Different Concentrations of Sucrose

Concentration of Sucrose (M)	Initial Mass (g)	Final Mass (g)	Change in Mass (g)	Percent Change in Mass (%)
0.0	25.48	25.56	0.08	0.313
0.2	25.47	26.56	1.09	4.10
0.4	28.25	31.03	2.78	8.96
0.6	27.45	30.54	3.09	10.1
0.8	27.97	32.08	4.11	12.8
1.0	28.37	33.66	5.29	15.7

```
% MATLAB script to plot diffusion flux at different sucrose concentrations
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Define data from Table 2
sucrose_conc = [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]; % Sucrose concentrations in M
initial_mass = [25.48, 25.47, 28.25, 27.45, 27.97, 28.37]; % Initial mass of bags in grams
final_mass = [25.56, 26.56, 31.03, 30.54, 32.08, 33.66]; % Final mass of bags in grams
initial_volume = 0.025; % Initial volume in liters (25 mL)
molar_mass_sucrose = 342.3; % Molar mass of sucrose in g/mol

% Calculate final volume (assuming water gain equals mass change in grams)
volume_change = (final_mass - initial_mass) / 1000; % Convert g to L
final_volume = initial_volume + volume_change;

% Calculate initial and final molarities
initial_molarity = initial_mass / (initial_volume * molar_mass_sucrose);
final_molarity = final_mass / (final_volume * molar_mass_sucrose); % Adjusted for final volume

% Compute concentration change (Delta C)
delta_c = initial_molarity - final_molarity;

% Define constants
delta_x = 0.00005; % Membrane thickness in meters
D = 5.22e-10; % Diffusion coefficient for sucrose in water (m^2/s)

% Calculate diffusion flux (J)
flux_j = -D * delta_c / delta_x; % Fick's First Law of Diffusion

% Plot results
plot(sucrose_conc, flux_j, '-o');

% Label the plot
xlabel('Sucrose Concentration (M)', 'FontSize', 14);
ylabel('Diffusion Flux (mol/m^2/s)', 'FontSize', 14);
title('Diffusion Flux vs. Sucrose Concentration', 'FontSize', 16);
grid on;
```

Figure 1. MATLAB code for a visual representation of the relationship between concentration of sucrose and diffusion flux to demonstrate the effects of osmosis in varying concentrations of sucrose.

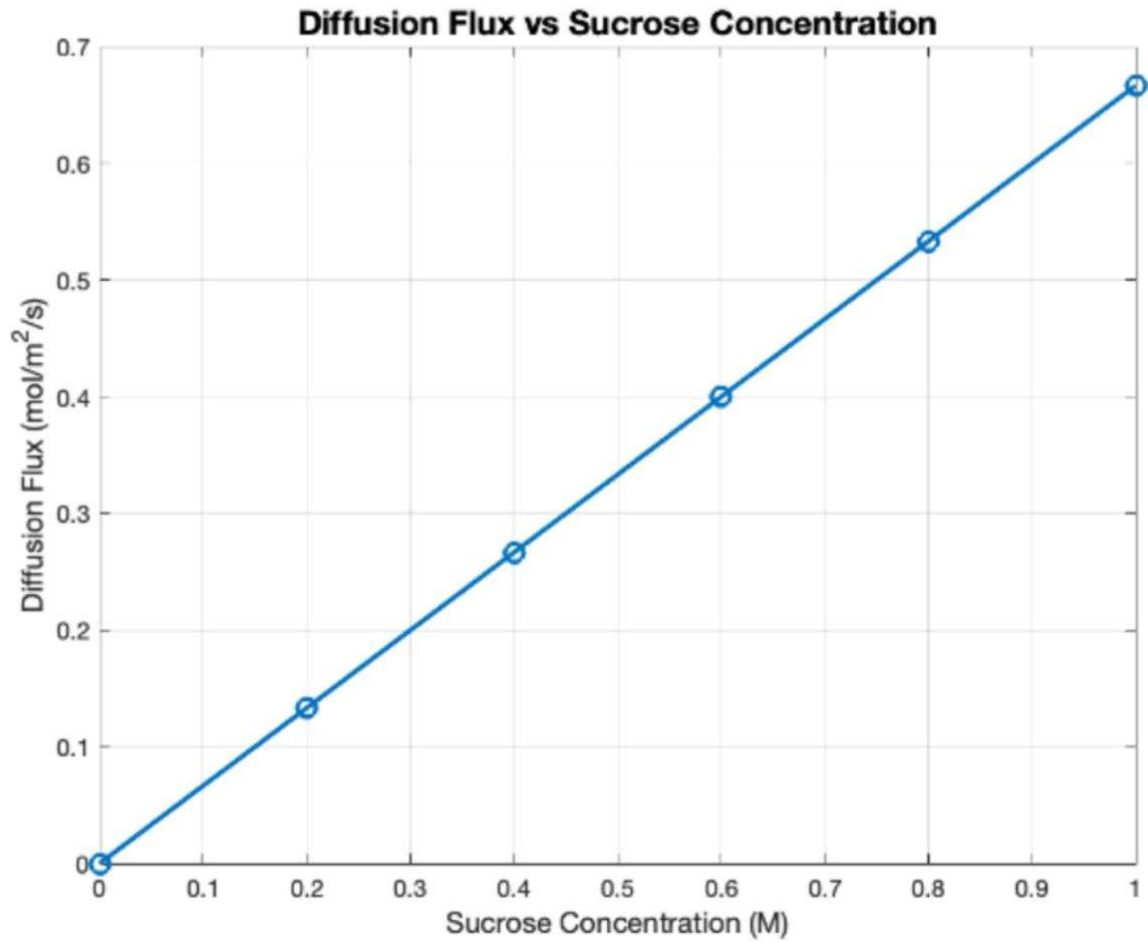


Figure 2. Representation of the linear relationship between concentration of sucrose and diffusion flux to validate Fick's First Law and to demonstrate the principles of osmosis and selective permeability.

Discussion

This experiment aimed to analyze the changes of osmolarity on different concentrations of sucrose to highlight its importance to the normal homeostasis of cells. Based on fluid dynamic principles, it was hypothesized that only iodine and glucose will diffuse through the tubing while starch will not, and as the concentration of sucrose in the solution increases, the change in mass and percent change in mass increases as well. The results of the experiment support the hypothesis, demonstrating a color change and positive glucose tests outside the bag, and a positive correlation between concentration of sucrose and change in mass/percent change in mass.

Table 1 displays the results of the solution color and glucose test results for glucose solution in dialysis bag and potassium iodide in a cup. The 15 % glucose / 1 % starch solution in the dialysis bag had no color and 2000 mmol/L of glucose initially. In the end the solution had a dark red color and 1000 mmol/L of glucose. The potassium iodide solution in the cup had a red/orange color throughout the experiment, and maintained 100 mmol/L of glucose as well. Therefore, the change in color and positive glucose test outside the bag reveals that iodine and glucose diffused through the tubing while starch did not.

Table 2 displays the results of the change in mass and percent change in mass for various concentrations of sucrose. The 0.0M concentration of sucrose had an initial mass of 25.48 grams, a final mass of 25.56 grams, a change in mass of 0.08 grams, and a percent change in mass of 0.313 %. The 0.2M concentration of sucrose had an initial mass of 25.47 grams, a final mass of 26.56 grams, a change in mass of 1.09 grams, and a percent change in mass of 4.10 %. The 0.4M concentration of sucrose had an initial mass of 28.25 grams, a final mass of 31.03 grams, a change in mass of 2.78 grams, and a percent change in mass of 8.96 %. The 0.6M concentration of sucrose had an initial mass of 27.45 grams, a final mass of 30.54 grams, a change in mass of 3.09 grams, and a percent change in mass of 10.1 %. The 0.8M concentration of sucrose had an initial mass of 27.97 grams, a final mass of 32.08 grams, a change in mass of 4.11 grams, and a percent change in mass of 12.8 %. The 1.0M concentration of sucrose had an initial mass of 28.37 grams, a final mass of 33.66 grams, a change in mass of 5.29 grams, and a percent change in mass of 15.7 %. Therefore, as the concentration of sucrose in the solution increases, the change in mass and percent change in mass increases as well, which is a result from an increase in osmotic gradient.

Furthermore, the relationship between concentration of sucrose and diffusion flux was illustrated in Figure 2. A line graph was produced showed that the relationship between the two variables was negative and linear. In the graph, the highest point of diffusion flux was at 1.0M concentration of sucrose, and the lowest point of diffusion flux was at 0.0M concentration of sucrose. This behavior aligns with theoretical expectations where diffusion flux is proportional to the concentration of sucrose.

The results obtained throughout the experiment are very similar to the values obtained by other groups performing the same experiment. Comparing values, the 1.0M concentration of sucrose was the highest change in mass/percent change in mass and highest diffusion flux, while 0.0M concentration of sucrose was the lowest change in mass/percent change in mass and lowest diffusion flux. Therefore, it can be concluded that the experiment is justifiable and accurate. To refine these findings, by understanding osmosis and diffusion, designs and modification to materials can be made to improve medical implants, drug delivery systems, and diagnostics devices, which is important for developing safer and more effective biomedical applications.

bottles provided in the kit. Then 160-170 mL of water were poured into 6 different cups, one for each solution, including 0.0 M and 1.0 M sucrose solutions. Once the bags were soaked for the allotted time period a piece was obtained from the soaked water. Then 25 mL of each sucrose solution was poured into the six different dialysis bags, with one end closed. Once the bag had the solution in it it was closed at the other end and weighed on the scale to obtain the weight, this value was recorded in grams. This was conducted for all the different solutions. Once this was conducted, the bags were placed in the cups of water for 30 minutes to allow for osmosis to occur. After 30 minutes the bags were removed from the solution and their weight was recorded.

Calculating the Change of Mass within the bag: Part B

For this experiment, the change of mass was calculated using equation 2.

$$\Delta m = m_f - m_i \quad (\text{Equation 2})$$

- Δm is the change in mass
- m_f is the final mass of the dialysis bag after soaking
- m_i is the initial mass of the dialysis bag before soaking

Results

Part A

In this part of the experiment, the diffusion was recorded and the hypothesis was that the smaller ions (Iodide) would diffuse into the dialysis bag while the larger solutes (starch) would not be able to diffuse out of the dialysis bag. This result did coincide with as shown in table 1 below.

Table 1. Results from Part A of the experiment

Location	Solution	Solution Color		Glucose Test Results	
		Initial	Final	Initial	Final
Dialysis Bag	Glucose/Starch	Clear	Beat Red	+++	+++
Cup	IKI	Burnt Orange	Sunset Orange (Lighter than initial)	++	++

The results show that the glucose/starch solution did not change over the 30 minute time period that the dialysis bag was in the iodine potassium iodine solution. This is known as the glucose test results did not differ from the initial to the final results. However, the smaller ions, the IKI did diffuse as shown in the dialysis bag containing the glucose/starch solution dialysis bag's change of color. This indicated that the dialysis bag allowed for diffusion across its membrane for the iodine particles, while it did not allow for the glucose/starch particles to diffuse out of the bag.

Part B

For this part of the experiment the hypothesis was that osmosis would cause the bags to increase in mass. This increase should be directly proportional with the concentration of sucrose in the dialysis bag. **Table 2.** and **Table 3.** show the change in mass of the dialysis bags for our own group and our change in mass compared to the other groups change in mass, respectively.

Table 2: Change of mass for our group

Contents In Dialysis Bag (M)	Initial Mass (g)	Final Mass (g)	Change in Mass (g)	Percent Change in Mass
0.0 M Sucrose (Distilled Water)	26.02	25.81	-0.21	-0.81%
0.2 M Sucrose	25.18	25.31	0.13	0.52%
0.4 M Sucrose	26.03	28.87	2.84	10.9%
0.6 M Sucrose	27.07	30.21	3.14	11.6%
0.8 M Sucrose	27.17	31.66	4.49	16.5%
1.0 M Sucrose	28.32	34.18	5.86	20.7%
Average	26.63 ± 1.1	29.34 ± 3.4	2.71 ± 2.3	10.2%

Table 3: Change of mass compared to other groups

Solution in Bag	Percent Mass Change			Total	Class Average
	Group 1	Group 2	Group 3		
0.0 M Sucrose	-0.81%	33.1%	0.00%	32.29%	10.76%
0.2 M Sucrose	0.52%	4.10%	7.50%	12.12%	4.04%
0.4 M Sucrose	10.9%	8.96%	12.10%	31.97%	10.66%
0.6 M Sucrose	11.6%	10.1%	14.70%	36.40%	12.13%
0.8 M Sucrose	16.5%	12.8%	20.40%	49.73%	16.58%
1.0 M Sucrose	20.7%	15.7%	24.70%	61.09%	20.36%

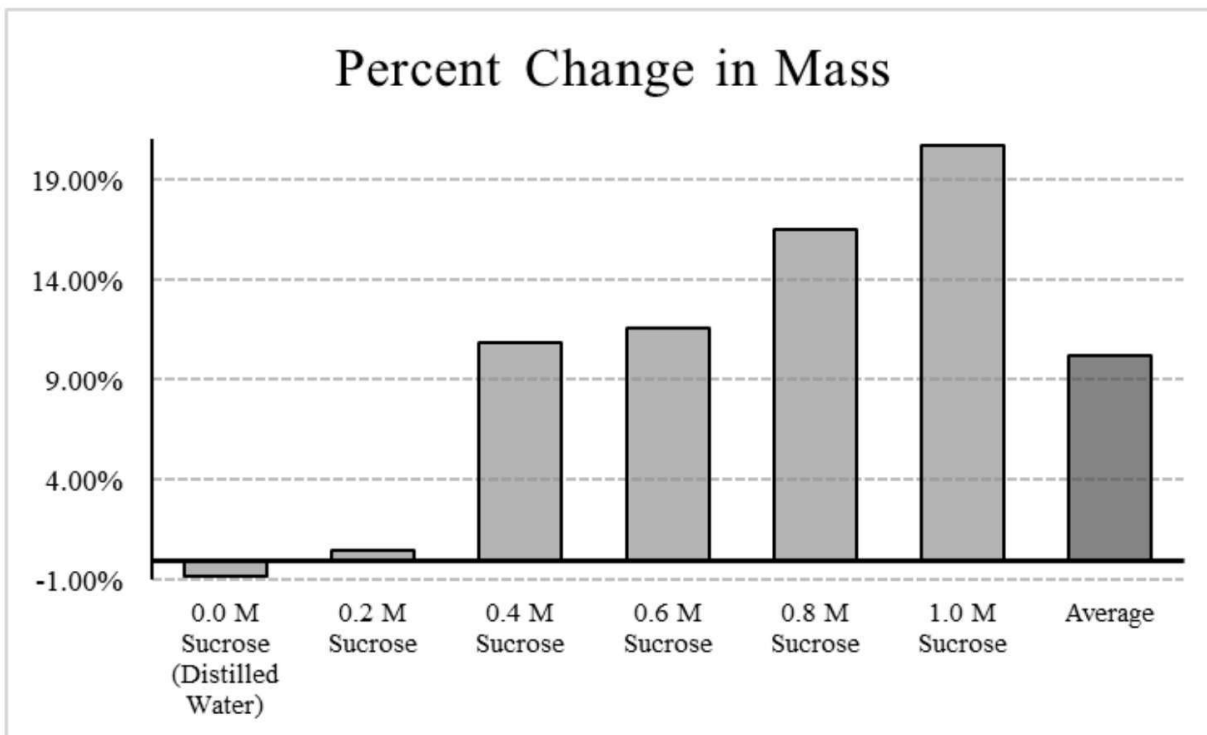


Figure 2: This figure graphically shows the results from **Table 2**, depicting our results as the sucrose solution increased (x-axis) and the percent change in mass increased (y-axis).

*This graph was created using Microsoft Excel.

From **Figure 2**, one can observe that as the solution concentration increases the percent change in mass also increases. For this experiment we also calculated the diffusion flux using Ficks first law equation.

$$J = -D \frac{\delta C}{\delta X} \quad (\text{Equation 3})$$

- J is the diffusion flux
- D is the diffusion coefficient
- δC is the change in concentration
- δX is the change in position

Using Fick's first law, Matlab code was written to compute the values. The code is as follows below:

```
% Define the given data
concentrations = [0.0 0.2 0.4 0.6 0.8 1.0]; % Molarity (M)
mass_change = [-0.21 0.13 2.84 3.14 4.49 5.86 ]; % Mass change (g)
position = 0.1; % Assumed membrane thickness (cm)
% Convert mass change to concentration change (approximation)
% Assuming volume remains constant, and molar mass of sucrose = 342.3 g/mol
volume = 0.1; % Assumed volume in L
delta_C = (mass_change ./ 342.3) / volume; % Molar concentration change (mol/L)
% Define the diffusion coefficient
D = 1.95e-5; % cm^2/s
% Compute diffusion flux using Fick's First Law
J = -D * (delta_C / position);
% Plot the results
figure;
plot(concentrations, J, '-o', 'LineWidth', 2, 'MarkerSize', 8);
xlabel('Sucrose Concentration (M)');
ylabel('Diffusion Flux (mol/cm^2·s)');
title('Diffusion Flux vs. Sucrose Concentration');
grid on;
```

The results produced from the Matlab Code yielded:

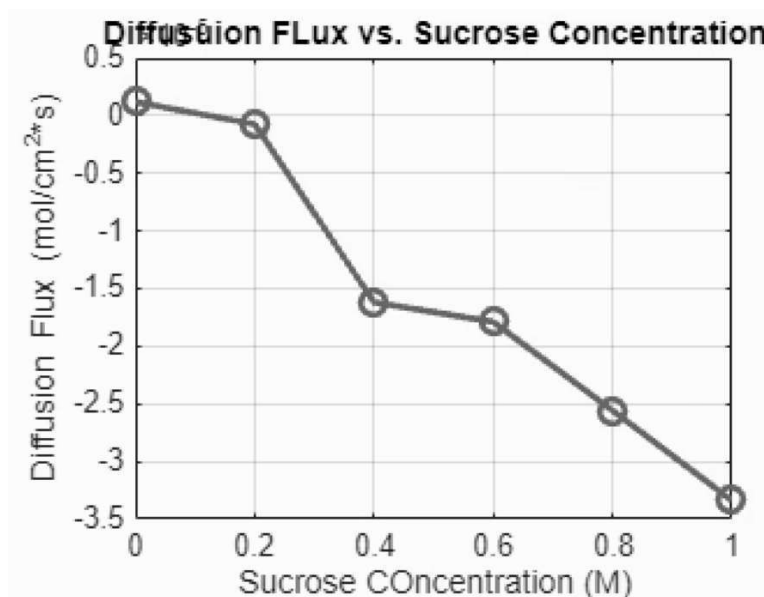


Figure 3. The graph was created using Matlab for Fick's first law of diffusion in a single direction.

Discussion

In this experiment for part A, the hypothesis was that the smaller ions (Iodide) would diffuse into the dialysis bag while the larger solutes (starch) would not be able to diffuse out of the dialysis bag. This hypothesis was supported from the results obtained in **Table 1**. from the color change and the glucose test strip results. This showed that the larger particles, glucose/starch solution, would not diffuse through the membrane of the dialysis bag. While the smaller particles, iodine solution, did diffuse through the interstitial fluid and the innards of the dialysis bag. For part B of the experiment the hypothesis was that osmosis would cause the bags to increase in mass. This increase should be directly proportional with the concentration of sucrose in the dialysis bag. From the Matlab code, the results showed a directly proportional, but not exactly, relationship between the concentration of the solution in the bag and the diffusion flux.

In the paper titled *Understanding and extending the Starling principle*, written by Michel CC, et al., they say that “because microvascular walls are permeable to plasma proteins, there are no true Starling equilibria but steady states are maintained by low levels of filtration in most tissues”, this paper hints that there must be more going on within the scope of the Starling principle and an extension of this principle is due for further comprehension. However, this paper does relate to the experiment conducted in a few ways. Firstly it relates with the oncotic pressure and osmolality. It essentially emphasizes the way that oncotic pressure influences fluid movement. In our experiment with the dialysis bags, the osmolality differences between the interstitial fluid and the solution on the inside of the bag drive water and solute diffusion. This is very similar to the main idea of the Starling principle in its influence under pressures for fluid moving across the capillaries. Another way it relates is that the paper highlights solutes and water moving across a semipermeable barrier, in the body it's the capillaries, while considering the factors of permeability and pressure gradients. In this experiment the dialysis bag imitates the features of permeability and the differing pressure gradients, in which the diffusion occurs due to these two factors. The third way the paper relates to the experiment is by discussing how fluid exchange essentially reaches a steady state that depends on the filtration and reabsorption processes. In the experiment, the diffusion will continue until the equilibrium concentration has been reached, at which point the osmolality between the interstitial fluid and the innards of the dialysis bags are equivalent.

Conclusion

In conclusion the goal of this experiment was to observe the effect of diffusion and osmolality across a semipermeable membrane. Understanding the importance of pressure

Results
Present data in appropriate form(Means and SD with correct units). Include Error bars if required. Use MATLAB code to interpret the data.
Results are expressed in text, and information is synthesized and organized logically according to the key topics.

Discussion
Provides a rational explanation of data within the context of relevant underlying concepts, speculates on the significance and relevance of results without overinterpretation, and discusses topics relating back to the results and significance to the experiments.
Applies information from additional sources to explain results.

DOES NOT MEET

Results

Activity A: Diffusion

The results of the diffusion experiment demonstrated the movement of solutes across the dialysis membrane and confirmed selective permeability. Table 1 summarizes the observations of solution color changes and glucose test results for both the dialysis bag and the surrounding cup. Initially, the glucose/starch solution inside the dialysis bag was clear, while the iodine potassium iodide (IKI) solution in the cup was orange/red. After 30 minutes, the solution inside the dialysis bag turned dark purple, indicating diffusion of iodine into the bag and its reaction with starch to form a starch-iodine complex. Meanwhile, the IKI solution in the cup remained orange.

Glucose test results further validated diffusion across the membrane. Initially, glucose was present only inside the dialysis bag (+), while it was absent in the surrounding cup (-). After incubation, glucose diffused out of the bag into the cup, as evidenced by positive glucose test results (++) . These findings highlight that small molecules like glucose can pass through the dialysis membrane, whereas larger molecules like starch remain confined within.

Table 1. Observation of Diffusion and Osmosis Using a Dialysis Bag

		Solution color		Glucose Test Results	
Location	Solution	Initial	Final	Initial	Final
Dialysis Bag	Glucose/ Starch	Clear	Dark Purple	+	+
Cup	IKI	Orange/Red	Orange	-	++

Activity B: Osmosis

The osmosis experiment revealed a clear relationship between sucrose concentration and water movement across the dialysis membrane. Table 2 and Figure 1 summarize changes in mass for dialysis bags containing varying sucrose concentrations. The percent change in

mass increased with sucrose concentration, indicating greater osmotic water influx into bags containing higher solute concentrations.

- At 0.0 M sucrose (distilled water), minimal change in mass was observed (0.17% class average), as there was no osmotic gradient.
- Bags with 0.2 M sucrose experienced a modest increase in mass (4.61%), while those with 0.4 M sucrose showed a larger increase (10.61%).
- Higher sucrose concentrations — 0.6 M, 0.8 M, and 1.0 M — resulted in progressively greater mass changes of 15.57%, 16.65%, and 24.95%, respectively.

These results align with osmotic principles: water moves from regions of lower solute concentration to regions of higher solute concentration to equalize osmolarity.

Table 2. Percent Change in Mass of Dialysis Bags at Different Sucrose Concentrations

Solution in Bag	Percent Change in Mass of Dialysis Bags (%)			Total (%)	Class Average (%)
	Group 1	Group 2	Group 3		
0.0 M Sucrose	0.08	0.12	0.313	0.513	0.171
0.2 M Sucrose	6.00	3.72	4.10	13.82	4.61
0.4 M Sucrose	13.82	9.06	8.96	31.84	10.61
0.6 M Sucrose	20.65	15.95	10.1	46.70	15.57
0.8 M Sucrose	22.76	14.39	12.8	49.95	16.65
1.0 M Sucrose	36.02	23.12	15.7	74.84	24.95

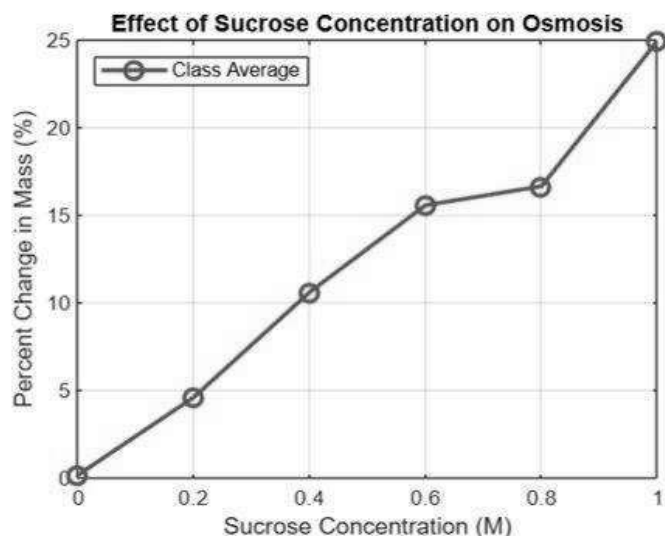


Figure 2. Percent Change in Mass as Sucrose Concentration Increases

Additionally, diffusion flux calculations based on Fick's First Law were performed using MATLAB to model water movement across membranes at different sucrose concentrations (Figure 3). The diffusion flux (J) decreased linearly as sucrose concentration increased, reflecting a steeper concentration gradient driving water movement into the dialysis bags:

$$J = -D \cdot \frac{\Delta C}{\Delta x}$$

Here, $D = 2.3 \times 10^{-9} \text{ m}^2/\text{s}$ is the diffusion coefficient of water, $\Delta x = 0.001 \text{ m}$, is membrane thickness, and ΔC represent concentration differences across the membrane. This computational analysis complements experimental findings by quantifying flux rates under varying osmotic conditions.

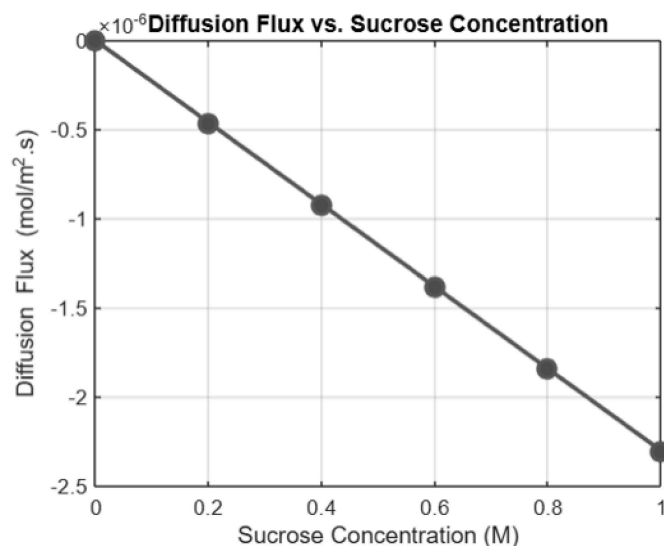


Figure 3. Water movement across membranes at different sucrose concentrations

Discussion

This experiment successfully demonstrated the fundamental principles of diffusion and osmosis, highlighting the selective permeability of membranes and the driving forces behind solute and water movement. The results from Activity A confirmed that small molecules like glucose can diffuse across a dialysis membrane, while larger molecules like starch remain confined. The color change inside the dialysis bag (from clear to dark purple) provided visual evidence of iodine diffusion into the bag and its reaction with starch, forming a starch-iodine complex. Additionally, glucose test results showed that glucose diffused out of the bag into the surrounding solution, consistent with its smaller molecular size and ability to pass through the membrane.

In Activity B, the relationship between sucrose concentration and osmotic water movement was observed. Dialysis bags containing higher sucrose concentrations exhibited greater increases in mass due to water influx driven by osmotic gradients. The percent change in mass increased proportionally with sucrose concentration, demonstrating that water moves

from areas of lower solute concentration (distilled water) to areas of higher solute concentration (sucrose solutions) to equalize osmolarity. These findings align with theoretical predictions based on osmotic pressure and Fick's First Law, as computational modeling further quantified diffusion flux rates under varying osmotic conditions.

The results of this lab have important physiological implications. In biological systems, similar mechanisms regulate water and solute movement across cell membranes and capillary walls. For example, osmosis is critical for maintaining cellular hydration, while diffusion governs nutrient and waste exchange in tissues. The principles demonstrated here also parallel the Starling equation for fluid exchange in capillaries, where hydrostatic and oncotic pressures drive fluid movement.

In conclusion, this experiment reinforced core concepts of diffusion and osmosis while providing quantitative insights into their dependence on solute concentration gradients. The findings highlight the relevance of these processes in both cellular and systemic physiology, emphasizing their role in maintaining homeostasis.

References

1. Lab Manual "Osmotic Properties of Human Red Blood Cells" by Promila Dhar, PhD
2. Michel, C. C., Woodcock, T. E., & Curry, F. E. (2020). "Understanding and extending the Starling principle". *Acta Anaesthesiologica Scandinavica*, 64(8), 1032–1037.
<https://doi.org/10.1111/aas.13603>

Comments from Instructor: Analysing results using MATLAB is missing

The discussion and conclusion could be more relevant and meaningful.